

Distributed LLM / RAG Load Balancing — sample knowledge base.

This document describes a simulated system with clients, a scheduler, a load balancer, and multiple GPU workers. Routing strategies include round robin (fair rotation), least connections (prefer the worker with fewest active sessions), and load-aware routing (prefer workers under a configurable connection threshold).

Sticky sessions bind each `client_id` to a worker for conversation context until that worker fails; failed requests retry on alternate workers up to `max_retries`.

Metrics collected include average and p95 latency, throughput in requests per second, failed requests, per-worker request counts, and simulated GPU utilization derived from busy time versus elapsed wall time per worker capacity.